

Class 1: Date: 16-Jan-2026

[Lecture Sheet 1]

Qs 1: Define Signal

Ans: Slide 3: Para 1 + Diagram

Qs 2: How a Signal is transformed to frequency domain?

Ans: Slide 4: Para 1 + Diagram

Qs 3: Compare discrete time signal and digital signal.

Ans: Slide 8: Last Para + Diagram + Slide 9 Full

Qs 4: Draw the complete diagram to convert the analog signal to digital signal.

Ans: Slide 13 Full

Qs 5: State Nyquist sampling criteria.

Ans: Slide 17

Qs 6: Example:

An analog signal of highest frequency of $B = 3.4$ KHz sampled maintaining a guard band of 1200Hz, find sampling rate f_c and sampling period T .

Ans: Slide 18 (Draw the Spectrum + Math)

Class 2: Date: 24-Jan-2026 (Online)

[Lecture Sheet 1]

Q 1: How Quantization is done and what is its impact on received Signal?

Ans: Slide 19

Q 2: Show PCM technique with diagram.

Ans: Slide 20

Q 3: Give the concept of DPCM and Delta modulation with Diagram.

Ans: Slide 25+26

Q4: Solve the problem on Line Coding.

Ans: Slide 31

[CT1 Syllabus End here]

Q 5: Define Multiplexing and Multiple access.

Ans: Slide 41

Q 6: Give the concept of Frequency Division Multiplexing (FDM).

Ans: Slide 43 (Diagram + Para 1)

Q 7: Give the concept of Time Division Multiplexing (TDM).

Ans: Slide 44 (Yellow Para + Diagram 1)

[Lecture Sheet 1 End Here]

[Lecture Sheet 2 Start Here]

Q 8: What is the function of Modem in a computer network?

Ans: Slide 1 (Para 2)

Q 9: Why is digital modulation necessary in wired communication?

Ans: Slide 2: Any 2 points

Q 10: Why is digital modulation necessary in wireless communication??

Ans: Slide 3: Any 3 points

Class 3: Date: 30-Jan-2026

[Lecture Sheet 2]

Q 1: Write basic principles of BFSK.

Ans: Slide 4: Shaded Part

Q 2: Draw binary FSK modulator circuit.

Ans: Slide 5: Diagram

Q 3: Draw FSK demodulator circuit.

Ans: Slide 6: Diagram

Q 4: Write basic principles of BPSK.

Ans: Slide 8 :Full

Q 5: Why M-ARY PSK is preferable compared to BPSK? .

Ans: Slide-11 (Para1 - Shaded Part)

Q 6: Derive the initial phase angles of 8-PSK?

Ans: Slide-11 (Last 2 Line calculation)

Q 7: Why Quadrature Amplitude Modulation is widely used in data communication?

Give the expression of modulated wave & draw a simplified modulated wave (draw shape of QAM).

Ans: Second equation-টি বাদ→ Slide-18 + Diagram → Slide-19

Class 4: Date: 06-Feb-2026

[Lecture Sheet 3 Start Here]

Q 1: Draw Three layer architecture of two philosophers communication.

Ans: Slide 18

Q 2: Mention two basic functions of the Physical layer.

Ans: Slide 28 -> Point 2, 3

Q 3: Mention three Basic functions of Logical Link Control (LLC).

Ans: Slide 30 - Para 3 + Slide 31 - Para 2 + Slide 32 - Para 1

Q 4: Compare Error detection and error correction.

Ans: Slide 31 - Para 1 (Optional) + Para 2 + Para 3

Q 5: Compare Physical and Logical address.

Ans: Slide 36 - Para 1 + Para 2

Class 5: Date: 14-Feb-2026 (Online)

[Lecture Sheet 4 - Data Link Layer: Start Here]

Q 1: What is the function of the header and tail of a frame?

Ans: Slide 3 - Para 1

Q 2: Define frame with diagram.

Ans: Slide 2 - Para 1 + Diagram

Q 3: Show Bit stuffing with Diagram.

Ans: Slide 10 - Diagram

Q 4: Compare performance of Bit Stuffing and Byte Stuffing.

Ans: Slide 11 - Para Diagram

Q 5: Give the idea of 'error correction through retransmission' and 'autonomous error correction',

Ans: Slide 13 - Para 2+ Para 3

Q 6: Write the steps of the Cyclic Redundancy Check (CRC) algorithm.

Ans: Slide 17 - Paste Colour part.

Q 7: Example 1:

Ans: Slide 18 + Slide 19

Q 8: How to avoid-

- a) Single-bit error
- b) two bit error
- c) odd number of bits in error

Ans: Slide 24+Slide 25: Point 1, 2, 3

Q 9: What is the probability of CRC failing to detect error?

Ans: Slide 26 - Para 2

[CT 2 Syllabus End here]

Class 6: Date: 20-Feb-2026

[Lecture Sheet 4 - Data Link Layer]

Q 1: Define flow control of LLC Sub Layer.

Ans: Slide 28 - Para 3

Q 2: Give the basic working principle of Stop-and-Wait Flow Control.

Ans: Slide 29 - Full

Q 3: Draw few steps of Sliding Window Flow Control.

Ans: Slide 36, 37, 38 - (No Description only diagram)

Q 4: What is the main problem of random access protocol?

Ans: Slide 46 - Para 3 (except 1st and Last Sentence)

Q 5: What is the main drawback of Controlled access protocol?

Ans: Slide 47- Para 2

Q 6: Compare *Exhaustive Service* and *Gated Service* of Token Ring Protocol.

Ans: Slide 49, 50 - (a) & (b)

Q 7: Why is CSMA/CA used in wireless network?

Ans: Slide 53 - Para 1

Class 7: Date: 27-Feb-2026

[Question given in next class]

[Lecture Sheet 5 - The Network Layer]

Class 8: Date: 6-Mar-2026 [CT only No Class conducted]

[Lecture Sheet 5 - The Network Layer]

Q 1: Draw a network with layered architecture which is heterogeneous at data link layer but homogeneous at network layer.

Ans: Slide 4 - Diagram Only

Q 2: Mention the function of Differentiated Services of IP Packet.

Ans: Slide 7- Description only - No Diagram

Q 3: Explain operation of IP Fragmentation and reassembly with Diagram.

Ans: Slide 10, 11 (Excluding the Table of Slide 11)

Q 4: Give the concept of Tunneling with appropriate diagram, considering connection of IPV6 routers through IPV4 routers.

Ans: Slide 23 - Diagram Only

Q 5: What is Subnet Mask? What is the function of Subnet Mask?

Ans: Slide 26 - (Excluding Last Para)

Q 6: Problem on SLIDE 34. Example-2

Ans: Slide 34

[CT 3 Syllabus Start here]

Class 9: Date: 14-Mar-2026

Lecture Sheet 6: Packet Routing

Q 1: Compare Datagram and Virtual Circuit

Ans: Slide 3 - Full

Q 2: Routing Table of Connectionless service. Graph to Table

Ans: Slide 6- Full [Network Diagram will be given - Have to prepare the Table]

Q 3: Routing Table of Connection-Oriented service. Graph to Table

Ans: Slide 9 - Full [Network Diagram will be given - Have to prepare the Table]

Q 4: Compare Virtual-Circuit and Datagram Subnets.

Ans: Slide 11- Any 3 Point

Q 5: Compare Adaptive and Non Adaptive algorithm

Ans: Slide 13-Last 2 Para

Q 6: Define fairness and Optimality of a Network.

Ans: Slide 14-Para 2, 3

Q 7: Conflict with efficiency and fairness:

Ans: Slide 15 - Full

Q 8: Problem on Sink Tree

Ans: Slide 18, 19, 20

Q 9: Which parameters are considered to determine the Path length in a Network?

Ans: Slide 21-6 point

Q 10: Mention Remedial Measure of Flooding in a Network.

Ans: Slide 33-Last 3 Para

[CT 3 Syllabus End here]

Class 10: Date: 28-Mar-2026

Lecture Sheet 7: Transport Layer

Q 1: Give the basic Concept of logical and Physical Communication.

Ans: Slide 3-Text Only - No Diagram

Q 2: Draw three-way handshaking protocol with appropriate diagram.

Ans: Slide 11 - Diagram Only - No Description

Q 3: How old duplicate CR is avoided in the transport layer?

Ans: Slide 12 - Diagram Only

Q 4: How old duplicate CR and old duplicate Data is avoided in the transport layer?

Ans: Slide 13 - Diagram Only

Q 5: Compare Asymmetric release and Symmetric release.

Ans: Slide 14 - Para 2

Q 6: Draw three-way handshaking for connection release.

Ans: Slide 18 - Diagram

Q 7: Give the basic concept of demultiplexing and multiplexing.

Ans: Slide 23 - Full

Q 8: Give the idea of the congestion window and the receive window.

Ans: Slide 28 - Para 2, 3, 4 (White part)

Q 9: Draw the profile of TCP Tahoe.

Ans: Slide 31- Diagram Only (Free hand drawing)

Q 10: Draw the profile of TCP Reno.

Ans: Slide 34 - Diagram Only (Free hand drawing)

Class 11: Date: 3-Apr-2026

Lecture Sheet 8: The Application Layer

Q 1: Compare client-server and P2P architecture at application layer.

Ans: Slide 2 - Para 2 + Slide 3 - Para 1

Q 2: Draw the arrangement of socket in process to process communication.

Ans: Slide 6 - Only Diagram - No Description.

Q 3: How a process is addressed in a host/network?

Ans: Slide 7- Full Slide

Q 4: Give the idea of network application and application protocol.

Ans: Slide 9 - Para 2

Q 5: Draw the hexarchy of domain name and labels with example.

Ans: Slide 15 - Only Diagram - No Description.

Q 6: Mention the function of three phases of POP3.

Ans: Slide 25 - Full

Class 12: Date: 10-Apr-2026

Lecture Sheet 8: The Application Layer

[No Question given]